

Discrete Groups & their reps:

$G \cong H$ iff $\exists$ homo $\varphi: G \rightarrow H$ which is bijective

(φ^{-1} exists) and $\varphi^{-1}: H \rightarrow G$ is also a group homo.

Automorphism: (or symmetry) of G is an isomorphism $G \rightarrow G$.

$\text{Aut}(G)$

ex: $G = \mathbb{Z}_n = \{[0], [1], \dots, [n-1]\}$

$$U_1(\mathbb{C}) \equiv \{z \in \mathbb{C} \mid |z| = 1\}$$

$$U_1(\mathbb{C}) \cong S^1$$

Now, define the homomorphism $\varphi: \mathbb{Z}_n \rightarrow U_1(\mathbb{C})$

$$\varphi([a]) := \exp(2\pi i a/n)$$

Proof: $\varphi([a] +_{\mathbb{Z}_n} [b]) = e^{\frac{2\pi i}{n}(a+b)} = e^{\frac{2\pi i}{n}a} e^{\frac{2\pi i}{n}b}$
 $= \varphi([a]) \cdot_{U_1(\mathbb{C})} \varphi([b])$

ex: $\{\mathbb{R}^+, +\} \cong \{\mathbb{R}^+, \cdot\}$

The isomorphism is given by the log.

Proof:

$$\log: \mathbb{R}_+^+ \rightarrow \mathbb{R}_+^+$$

$$\log(a \cdot b) = \log(a) + \log(b)$$

Here the image of $\log$ is $\mathbb{R}$ and not just $\mathbb{R}^+$.

ex: $\text{Aut}(\mathbb{Z}) \cong \mathbb{Z}_2$

$$[\text{Aut}(\mathbb{Z}) = \{\varphi: \mathbb{Z} \rightarrow \mathbb{Z} \mid \varphi \in \text{hom}(\mathbb{Z}, \mathbb{Z})\}]$$

Field: $\mathbb{C}, \mathbb{R}, \mathbb{Q}$

Characteristic 0 $\equiv 0$ is the only integer k w the property $kx = 0 \forall x$ in the field.

$$\rightarrow \mathbb{Z}_p: \{[0], \dots, [p-1]\}$$

multiplication by 0 has the same effect

as 0:

$$P[k] = \underbrace{k + \dots + k}_p$$

ex: $P=3; \mathbb{Z}_3 = \{[0], [1], [2]\}$

$k=1: \Rightarrow [3] = [0] \checkmark$

$k=2: [6] = [0]$

Any multiple of p is equal to "[0]".

Index: of a subgroup $H \subset G$ is the number of 'cosets' gH

For a finite group this is simply $|G|/|H|$

→ Normal subgroup: $N \subset G$ is one obeying $gNg^{-1} = N \quad \forall g \in G$

$G/H = \{[gH]\}$ (Set of cosets)
has a natural group structure when H is Normal.

* If H is a normal subgroup of G we write $H \triangleleft G$

* $\text{Ker}(\varphi) \equiv \{ \forall g \in G \mid \varphi(g) = e_H \}$
i.e.

$$\varphi(\text{Ker}(\varphi)) = e_H$$

$$\text{Im}(\varphi) \cong G/\text{Ker}(\varphi)$$

Free Group: P_n w/ generators $\{x_1, \dots, x_n\}$
is the set of all possible word in the 'alphabet'

$x_1, x_1^{-1}, \dots, x_n, x_n^{-1}$, w/ group operation given by concatenation

$$\text{ex: } F_1 \cong \mathbb{Z}, \quad \{P(1), +\}$$

$$\left[\begin{array}{l} \{1, 1-2=0, 1-2=-1, 1+1=2, \dots\} \\ \cong \mathbb{Z} \end{array} \right]$$